

**Dominance can
move when
ultimate-pair
activity moves.**



**Centre for Blockchain
Technologies**

The Making of Dominant Vehicle Currencies

Evidence from DeFi
Jiahua Xu

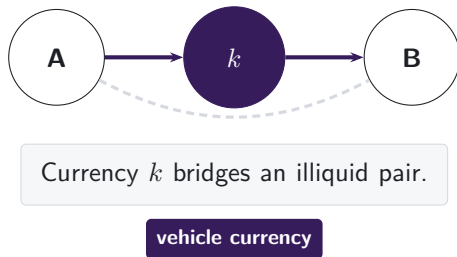
A cross-border payment needs someone to make the FX market

THE LIQUIDITY-PROVIDER PROBLEM

A payment provider receives one currency but must deliver another it does not hold.

Project Nexus requires an FX provider.
Project Rialto uses a vehicle when direct exchange is unavailable.

Note: Project Nexus states that every cross-border, cross-currency payment needs an actor willing and able to exchange the currencies. Project Rialto experimentally uses a vehicle currency when direct exchange is unavailable. The paper studies the intermediary asset observed inside DEX pool routes.

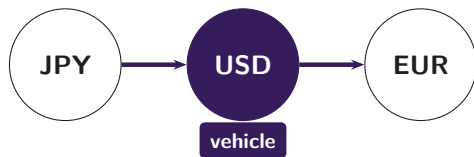


Pool routes reveal the vehicle currency

FX ANALOGY

Yen–euro may trade through dollars.

Conventional turnover data show two legs, not the customer's connected route.



Pool logs reveal the path.

DeFi is on-chain finance executed by smart contracts; decentralised exchanges (DEXs) trade tokens through liquidity pools. **Stablecoins** are tokens designed to track a reference currency, usually the US dollar.

Note: Traditional FX records show turnover in each currency pair, but not whether two legs belong to one customer route. The DEX setting preserves the transaction-level pool path, so the intermediary asset is observed directly.

The route panel links pool-level swaps

472 million

pool-level swaps

2,332

calendar dates

8

DEX deployments

February 2020–June 2026

Uniswap v2, v3, and v4; SushiSwap v2 and v3; Curve; Balancer; and Fluid.

Note: The listed deployments form the principal Ethereum route panel. Uniswap v1 enters the separate protocol-architecture analysis because its retained records do not recover endpoint-pair identities.

Pool data may start after the user instruction

USER INSTRUCTION RECORDED BY THE EXPLORER

Transaction details # 0x

⚡ Swap 460.00K  PYUSD for 460.10K  USDe on 0x  0x

THE DECISIVE SETTLEMENT TRANSFERS

 USDC
ERC-20 **Transfer**

 Market Maker: 0x51c7.....  →  Mainnetsettler 

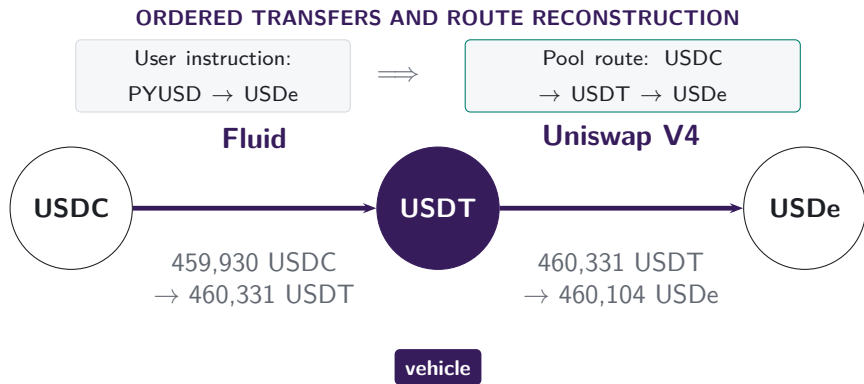
 USDC
ERC-20 **Transfer**

 Mainnetsettler  →  Fluid: Liquidity 

The instruction is PYUSD → USDe. The connected pool route begins with USDC.

Note: Transaction 0xbda1d07f...9bf67ad9, 10 January 2026. The data do not reveal whether PYUSD–USDC is another vehicle leg or executor inventory.

Inside the pools, USDT links USDC to USDe



The observed endpoint pair is USDC → USDe. Its legs are USDC → USDT and USDT → USDe; USDT is the vehicle that links them.

Note: The route is the connected pool-swap component of transaction 0xbda1d07f...9bf67ad9 on 10 January 2026.

Vehicle dominance is an intermediary share

MEASUREMENT

- Dominance is the share of indirect trade intermediated by asset k .
- Count dominance weights each route equally; value dominance weights comparable routed value.



Count-weighted dominance

Share of indirect routes that pass through k

Value-weighted dominance

Share of comparable routed value that passes through k

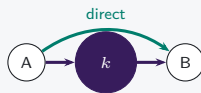
Architecture changes routing opportunities

V1 mandatory hub



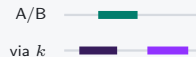
Hub use is imposed

V2 atomic-pair choice



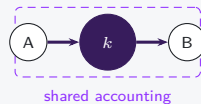
Hub use becomes a
choice

V3 range choice



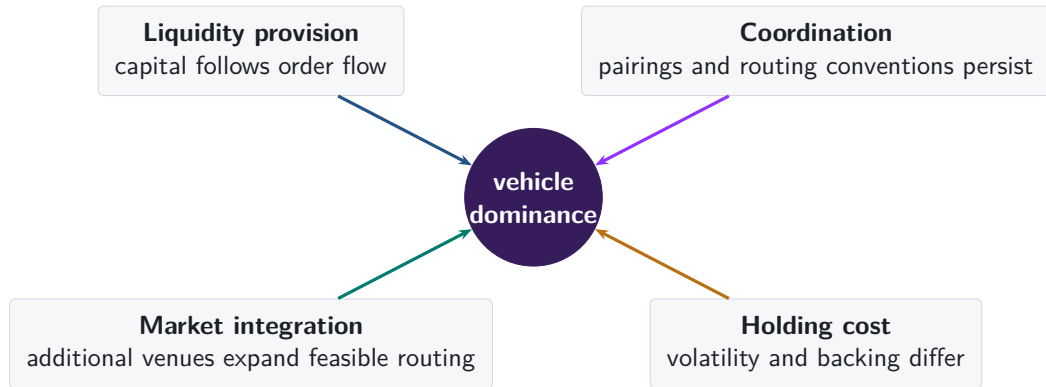
Atomic-pair
opportunities shift

V4 net settlement



Settlement differs
from route use

Four economic forces shape vehicle dominance



Native-asset pairing persists after it becomes optional

V1: PROTOCOL RULE



**217,003 token-to-token trades
pass through ETH**

Every pool is an ETH–token atomic pair,
so the architecture imposes the vehicle.

V2: MARKET CHOICE

95.5%

of single-leg V2 trades use a WETH
pool in 2026

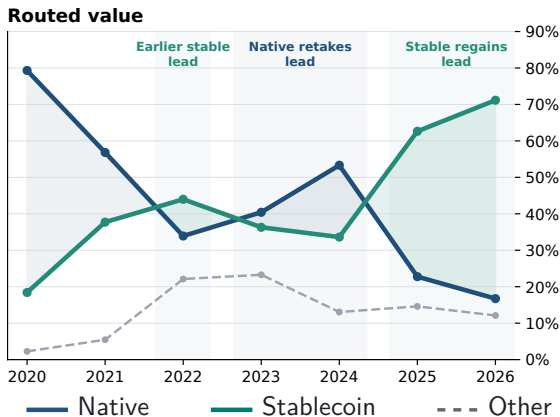
97.9%

of endpoint pairs first traded in 2026
include WETH

Note: V1 links token-to-ETH and ETH-to-token swaps within one transaction. V2 reports single-leg trades and newly traded endpoint pairs on that venue.

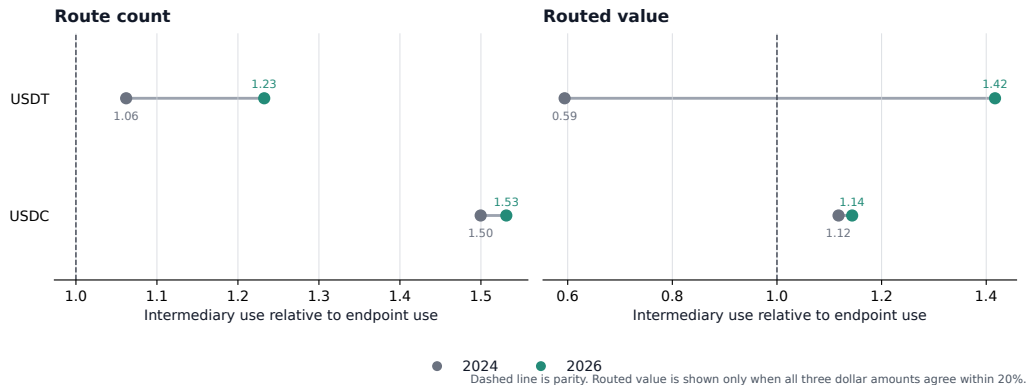
Stablecoins regain the routed-value lead by 2026

MATCHED JANUARY-JUNE PERIODS



Note: Exact two-leg routes. Routed value requires agreement among source, intermediary, and destination dollar values. The 2026 endpoint covers January through June.

USDT carries the stablecoin expansion



USDT crosses parity by routed value and extends its count advantage; USDC changes little.

Note: Excess use divides intermediary share by route-endpoint share on the same perimeter. The figure compares the same January through June dates in 2024 and 2026.

Endpoint demand predicts vehicle use within day

$$I_{a,t} = \alpha + \beta D_{a,t} + \mathbf{A}'_a \theta + \mu_t + \eta_a + \varepsilon_{a,t}$$

$I_{a,t}$ intermediary episode share

μ_t date effects

$D_{a,t}$ own endpoint-demand share

η_a currency effects

$\mathbf{A}_a$ asset-class indicators

Note: Outcome and demand are shares of the same day's route universe, measured in percentage points (pp). The estimates are descriptive within-day associations.

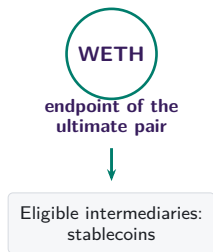
The demand relationship survives currency and date effects

	Intermediary share ($I_{a,t}$) [pp]			
	(1) Pooled	(2) Date FE	(3) + demand	(4) + currency FE
Native currency	+34.56 (23.04)	+34.55 (23.04)	+0.05 (0.56)	— —
Stablecoin	+2.42* (1.32)	+2.42* (1.32)	+0.85* (0.50)	— —
Endpoint-demand share ($D_{a,t}$)	— —	— —	+1.59*** (0.03)	+0.98*** (0.30)
Date fixed effects	No	Yes	Yes	Yes
Currency fixed effects	No	No	No	Yes

Endpoint-demand share passes through almost one for one after both fixed effects.

Note: 1,828,862 currency-days and 2,259 dates. Parentheses report currency/date-clustered standard errors. Class coefficients are percentage-point differences; the demand coefficient is percentage points of intermediary share per percentage point of endpoint-demand share. Asterisks *, **, and *** denote significance at 10%, 5%, and 1%, respectively.

WETH-linked endpoint pairs carry the count rotation



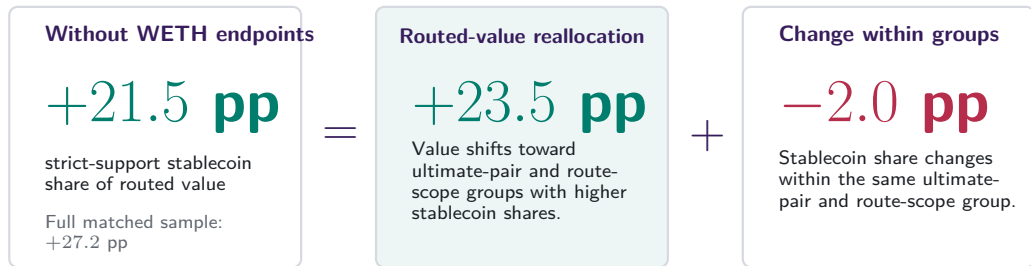
+21.1 pp All matched ultimate-pair groups
WETH-linked route-count mass: 26.6% → 46.8%

+3.7 pp Ultimate-pair groups excluding WETH endpoints
79,347 matched groups

Fixed-endpoint-pair changes are +0.2 pp overall and +0.3 pp without WETH endpoints.

Note: Exact two-atomic-trade native-WETH-versus-stablecoin routes on common month-days. Within-endpoint-pair estimates fix end-point pair, month-day, and route scope.

Trading shifts toward stablecoin-heavy endpoint pairs



Note: The daily midpoint identity is exact after removing WETH endpoints. Routed value requires agreement among source, intermediary, and destination dollar values.

New endpoint pairs deliver the largest positive component

EXACT POOLED DECOMPOSITION, 2024 TO 2026

2024 **16.9%** → 2026 **42.5%** **+25.7 pp**
aggregate rotation

Continuing endpoint pairs

−0.1 pp net within-endpoint-pair change

+8.6 pp activity reallocation

Endpoint-pair entry and exit

+21.0 pp newly traded endpoint pairs

−3.3 pp endpoint-pair exit

Pair entry contributes +21.0 pp; net switching inside continuing pairs contributes −0.1 pp.

Note: Matched January–June periods in 2024 and 2026. The within-pair component combines +1.3 pp toward stablecoins and −1.4 pp toward native assets. The remaining continuing-pair-weight component is −0.5 pp. Components sum using unrounded values.

Local bridge depth predicts route choice

84.1%

route share captured by
the deepest bridge

2024 **75.5%**

2026 **86.5%**

deepest-bridge share

+6.79 pp

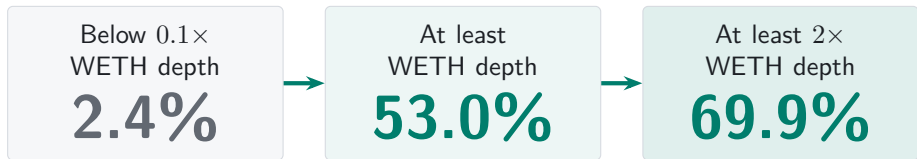
per log point of
weaker-leg deposited
capital

The weaker atomic leg carries the economically relevant depth constraint.

Note: The analysis compares five vehicle assets within locally supported endpoint-date-route-scope sets. Depth is $\log(1 + \text{prior deposited capital in USD})$ on the weaker atomic leg; the regression also controls for vehicle reach.

Stablecoin penetration rises with relative bridge depth

FIRST 30 DAYS AFTER STABLE-BRIDGE AVAILABILITY



Availability defines the route set; relative depth predicts penetration.

Note: 11,327 endpoint-pair days across 855 bridge events. Depth is prior-day deposited capital on the shallower atomic leg.

V4 internal routing is followed by LP repositioning

120-DAY RESPONSE TO 10 PP MORE INTERNAL SAME-ASSET ROUTING

$$Y_{c,t+120} = \beta M_{c,t} + \mathbf{X}'_{c,t}\gamma + \eta_c + \tau_t + \varepsilon_{c,t}$$

+0.265***

vehicle-linked TVL [log points]

s.e. 0.031

+0.071***

LP actions [log points]

s.e. 0.013

+2.89 **pp**

wide/very-wide action share

s.e. 0.35 pp

More capital remains linked to the vehicle while providers reposition across broader ranges.

Note: 2,015 V4 vehicle-days and 403 date clusters. Vehicle and origin-date effects; origin-day swap activity, current vehicle-token-side LP flow, vehicle-linked TVL, LP actions, and range composition. Standard errors cluster by origin date. Predictive association, not a causal effect of V4 adoption. Asterisks *, **, and *** denote significance at 10%, 5%, and 1%.

Entry, depth, and provider response form dominance

1 New endpoint pairs carry the rotation and tend to persist.
Entry contributes +21.0 pp; 120-day own-stable shares: USDC 88.3%, USDT 87.7%.

2 Local depth predicts route leadership.
Stable route share: 2.4% below $0.1\times$ WETH depth; 69.9% at least $2\times$.

3 V4 internal routing predicts provider repositioning.
At 120 days: $+0.265^{***}$ log points of vehicle-linked TVL and +2.89 pp in wide/very-wide action share.

Dominance forms through market entry, local capital, and provider adjustment.

Appendix map

Definitions and data

- A1 Route units and vehicle status
- A2 Asset types and backing regimes
- A3 Sample construction and exclusions

Reconstruction and validation

- A4 Exact transaction state
- A5 Protocol-specific state
- A6 Forced routes in V1

Robustness

- A7 Timing and venue-scope robustness
- A8 Count and value weighting
- A9 Observed route endpoints

Mechanisms

- A10 Capital, inventory, and depth
- A11 Coordination and integration
- A12–A13 References

A1. One reconstructed route is one unit



one route unit
two swap legs

- Each linked $i \rightarrow k \rightarrow o$ route is counted once.
- Asset k is classified as the intermediary on that route.
- Dominance is the share of route units or supported value carried by k .

How we reconstruct a route from one transaction

Direct swap

one transaction



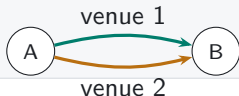
Sequential vehicle route

one transaction



Parallel split

one transaction



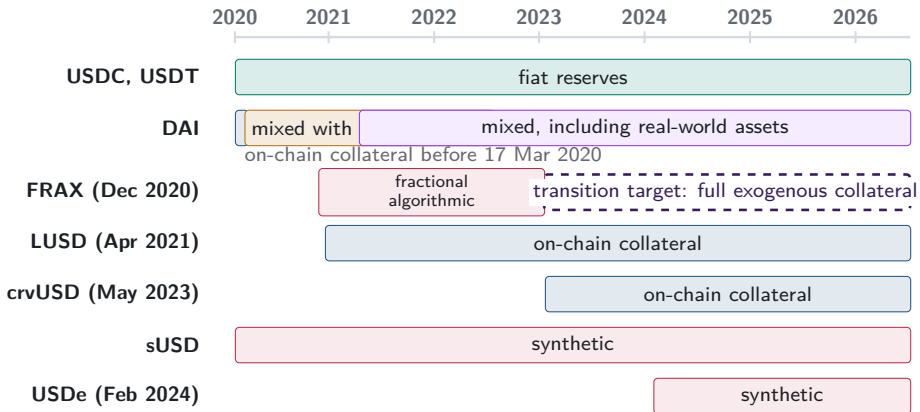
Round trip

one transaction



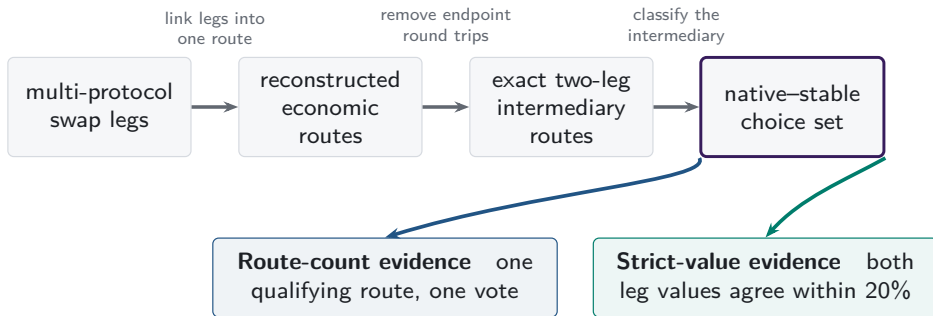
Vehicle choice directs indirect order flow toward particular assets and pools. Transaction-level paths make that choice visible inside aggregate turnover.

A2. Stablecoin backing changes over time



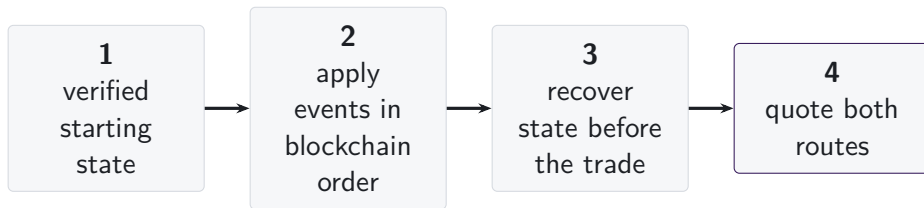
Note: Bars begin at public launch or the sample start for older tokens. DAI's dated regimes follow the Maker Governance actions that added USDC-A and activated RWA001-A. FRAX's dashed arrow follows FIP-188 and denotes a target, not observed collateral.

A3. One route universe supports two measurements



Atomic pairs inside multi-asset pools can supply route legs. V1/V2 architecture evidence is separate; dollar support changes value weighting.

A4. State is reconstructed immediately before execution



- Pool events are applied in blockchain order from a verified starting state.
- Each route is quoted using the pool state immediately before the transaction.
- Missing state intervals are excluded, and their share of trading volume is reported.

A5. Each AMM family has a different state vector

	reserves or active liquidity	ticks	amplification	weights or scaling	rates or oracles	fees	hook-dependent pricing
Constant product	■	—	—	—	—	■	—
Concentrated liquidity	■	■	—	—	—	■	—
StableSwap families	■	—	■	—	■	■	—
Weighted pools	■	—	—	■	■	■	—
V4 singleton	■	■	—	—	—	■	□

Executable-quote validation

■ recovered pricing input

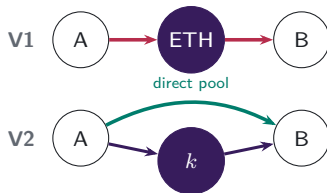
□ hook-dependent pricing unavailable

Protocol mechanisms studied

V4 singleton flash accounting and netting remain in scope; A5.3 shows the settlement mechanism.

A5.1. Atomic-pair creation determines available paths

V1 PROTOCOL RULE $\rightarrow$ V2 MARKET CHOICE



V1 admits only ETH-token atomic pairs. V2 permits any ERC-20 atomic pair, so market creation determines whether direct and vehicle paths coexist.

Which paths exist?

Economic implication: pool creation changes the feasible route set.

A5.2. Executable depth comes from active liquidity

V3 CONCENTRATED-LIQUIDITY POSITIONS



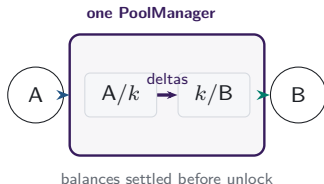
LPs choose a price range and fee tier for each atomic pool.
Positions covering the current price determine active liquidity
in the direct pool and each vehicle spoke.

Which path offers depth at this trade size?

Economic implication: executable depth is path- and size-specific.

A5.3. Shared accounting moves route settlement

V4 SINGLETON AND LOCK ACCOUNTING



One PoolManager holds all pools. During a lock, the two atomic trades update balance deltas; the caller settles its obligations before the lock ends.

Where is the route settled?

Economic implication: shared accounting changes the settlement boundary.

A6. V1 forced routes have an on-chain signature

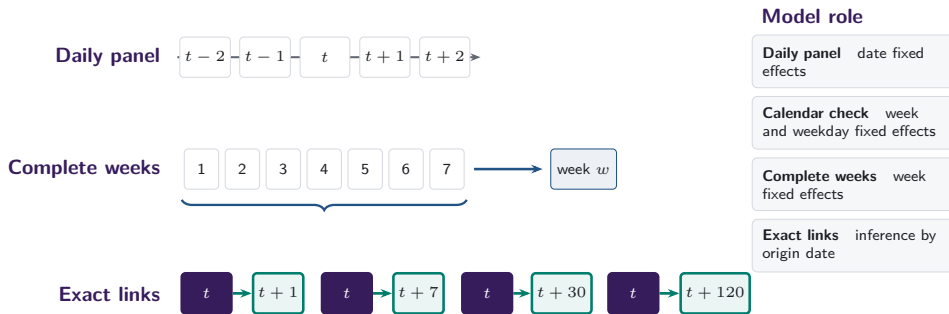
- Token-to-token execution emits two linked swaps through ETH.
- Matching ETH quantities distinguish protocol-imposed intermediation from unrelated bundled swaps.
- 129,810 mandate-era token-to-token transactions carry exactly matching ETH legs; the trace shows the largest.



same transaction, matching ETH flow

Note: Both legs of transaction 0x4dca160d...96a0ca16 (15 March 2020, block 9,674,728) report the same ETH amount to the last recorded digit.

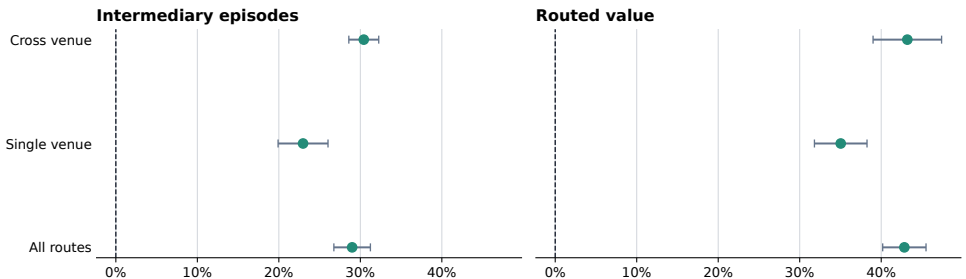
A7. Daily and weekly frequencies answer different questions



Exact horizons use the required target calendar date.

A7b. Stablecoin use rises within every venue scope

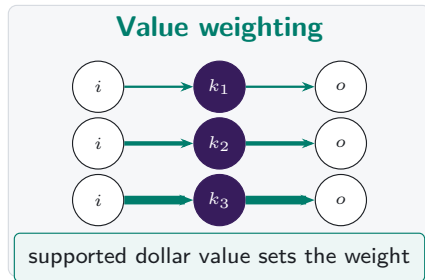
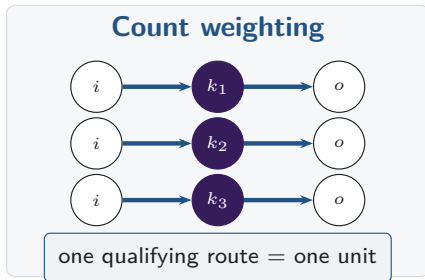
Stable share rises within realised route scopes



Points are 2024-to-2026 changes; bars are 95% HAC intervals.

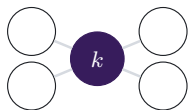
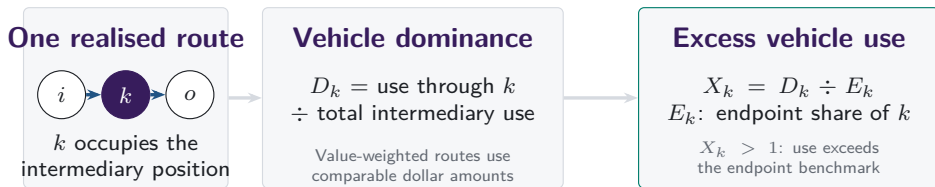
Note: Changes compare matched January-through-June dates in 2024 and 2026. Bars show calendar-HAC confidence intervals; venue scope records realised routes.

A8. Count versus value weighting

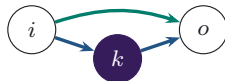


Note: Both panels contain the same routes. Line widths are illustrative, and value weighting applies only on the stated valuation-support band.

Vehicle dominance aggregates realised intermediary choices



Network position



Execution cost

Endpoint use is the benchmark; network position and same-state cost remain separate.

A9. Intermediary and route-endpoint use are separate

Intermediary share

$$I_{k,t}^N = \frac{n_{k,t}^I}{\sum_j n_{j,t}^I}$$

How often asset k sits inside a route.

Endpoint share

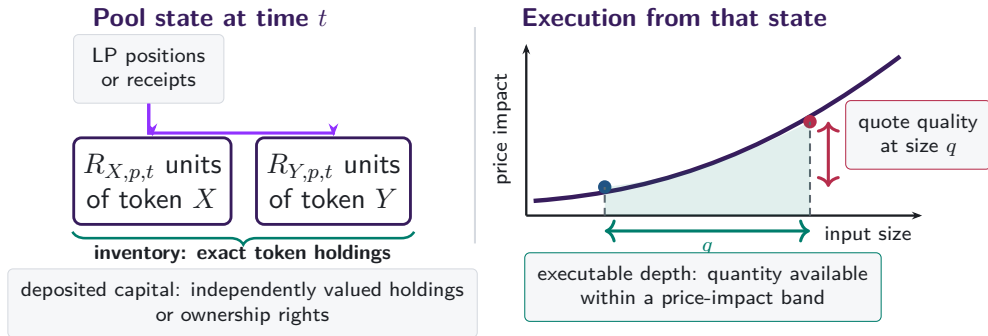
$$E_{k,t}^N = \frac{n_{k,t}^E}{\sum_j n_{j,t}^E}$$

How often asset k appears at either end of the observed pool route.

Observed pool-route endpoints supply the benchmark.

Excess use compares intermediation with observed route-endpoint use on the same support.

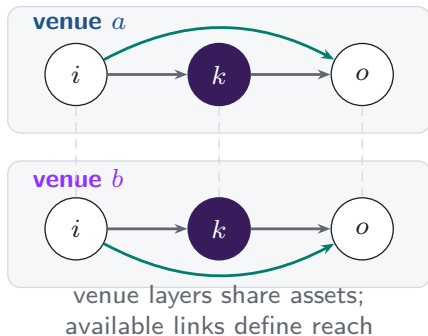
A10. Capital, inventory, and depth differ



Depth and quotes vary with size and state.

A11. Additional venues expand the feasible route set

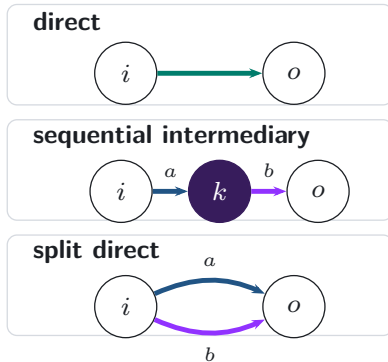
Feasible venue layers



realised
execution

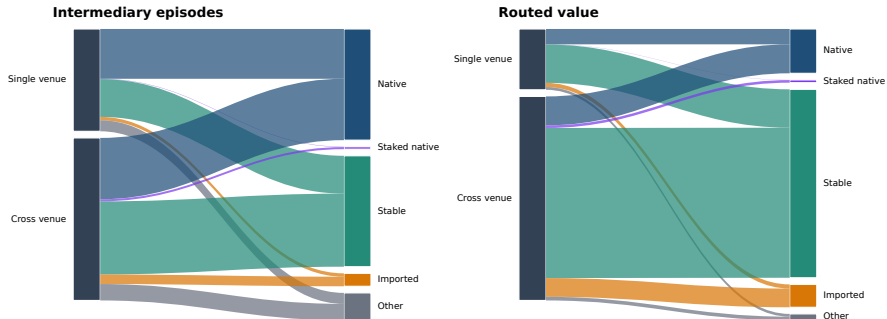


Realised route topology



A11b. Venue scope and vehicle type differ in 2026

Integration and vehicle composition are separate margins in 2026



Ribbon width is the share of 2026 intermediation jointly classified by venue scope and intermediary type.

Note: Ribbon width is the share of 2026 intermediation jointly classified by venue scope and intermediary type.

A12. References: vehicle currencies and market structure

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A13. References: decentralised exchange design

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